

Proof of Correctness: GS-MB (Margaritis & Thoun)

Assumptions: (1) Markov & Faithfulness
(2) Independence oracle

Grow Phase (to show): $\forall Y \in MB(X) : Y \in \mathbf{S}$ after the grow phase (or $MB(X) \subseteq \mathbf{S}$)

Proof by contradiction:

• Assume $\exists Y \in MB(X)$, for which $Y \notin \mathbf{S}$.

(i) $Y \in Pa(X)$ or $Y \in Ch(X) \Leftrightarrow Y$ is adjacent to X

by (1), (2) $Y \not\perp\!\!\!\perp X \mid \mathbf{S}$ (or any set not including $\{X, Y\}$) 

$\Rightarrow$ any Y in $Pa(X)$ or $Ch(X)$ will be added.

(ii) $Y \in MB(X)$ but $Y \notin Ch(X)$, $Y \notin Pa(X)$

Then, Y is a spouse of X and $\exists z : z \in Ch(X) \wedge z \in Ch(Y)$

a) $z \notin \mathbf{S}$, then by 1) it will be added

b) $z \in \mathbf{S}$, then z is a collider ($X \rightarrow z \leftarrow Y$)

and hence by (1)(2) $Y \not\perp\!\!\!\perp X \mid \mathbf{S}$ ($\Rightarrow$ add Y)

Hence, $MB(X) \subseteq \mathbf{S}$ after the grow phase.

Shrink Phase (to show)

S-1) No $\gamma \in MB(x)$ is removed from S

S-2) All $\gamma \notin MB(x)$ are removed from S

S-1) Similar to grow phase: Take a $\gamma \in MB(x)$ and assume γ is the first such element that we try to remove:

(i) $\gamma \in Pa(x)$ or $\gamma \in Ch(x) \stackrel{(1)(2)}{\Rightarrow} \gamma \not\perp x \mid S \setminus \{\gamma\}$

(ii) γ is a spouse & we know that a common child $z \in Ch(x) \cap Ch(\gamma)$ must be in $S \stackrel{(1)(2)}{\Rightarrow} \gamma \not\perp x \mid \underbrace{S \setminus \{\gamma\}}_{\text{includes collider } z \text{ between } x \text{ and } \gamma}$

S-2) Assume $\gamma \in S \wedge \gamma \notin MB(x)$.

• Let $Q = S \setminus (\{\gamma\} \cup MB(x))$ (all elements in S not in $MB(x)$ and not equal to γ)

• By the property of the Markov blanket:

$$x \perp\!\!\!\perp_S \{\gamma\} \cup Q \mid MB(x)$$

weak union

$(\Rightarrow)$

$$x \perp\!\!\!\perp_S \gamma \mid \underbrace{MB(x) \cup Q}_{= S \setminus \{\gamma\}}$$

$(1)(2)$

$(\Leftarrow)$

$$x \perp\!\!\!\perp \gamma \mid S \setminus \{\gamma\}$$

$\Rightarrow \gamma$ will be removed.

